

MATH 3312, SPRING 2026: PRACTICE PROBLEMS FOR THE FINAL

- (1) Fix an odd prime p and another prime $\ell \neq p$.
 (a) Show that $\mathbb{F}_{\ell^r}$ contains a zero for $\Phi_p(x)$ if and only if $\ell^r \equiv 1 \pmod{p}$.
 (b) Show that in $\mathbb{F}_{\ell}[x]$ we have

$$\Phi_p(x) = \prod_{i=1}^m f_i(x) \in \mathbb{F}_{\ell}[x],$$

where $(p-1)/m$ is the order of ℓ when viewed as an element of $(\mathbb{Z}/p\mathbb{Z})^\times$, and where $f_1(x), \dots, f_m(x) \in \mathbb{F}_{\ell}[x]$ are *distinct* irreducible polynomials of degree $(p-1)/m$.

- (c) Conclude that $\Phi_p(x)$ is irreducible in $\mathbb{F}_{\ell}[x]$ if and only if ℓ is a generator for $(\mathbb{Z}/p\mathbb{Z})^\times$.
- (2) Give an example of a square matrix A over some field k such that $\text{ch}_A(x) = x^3(x-1)^2$ and $\min_A(x) = x^2(x-1)^2$.
- (3) Use the Smith Normal Form of $A - xI_5 \in M_5(\mathbb{Q}[x])$ to compute the rational canonical form, the minimal polynomial, the characteristic polynomial, the eigenvalues and their algebraic and geometric multiplicities for the matrix

$$A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & -1 \\ 1 & -1 & 3 & 0 & 0 \\ 0 & 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 & 2 \end{bmatrix} \in M_{5 \times 5}(\mathbb{Q}).$$

- (4) For A in the previous problem, can you find a conjugate that is in Jordan canonical form?
- (5) If k is a field of characteristic 0, show that every irreducible polynomial $f(x) \in k[x]$ is separable.
- Definition 1.** A finite extension L/k is **separable** if, for every $\alpha \in L$, $m_\alpha(x) \in k[x]$ is separable.
- (6) Show that every finite separable extension L/k is contained in a Galois extension.
- (7) Write down all subextensions of $\mathbb{Q}(\zeta_5)/\mathbb{Q}$ and $\mathbb{Q}(\zeta_7)/\mathbb{Q}$.
- (8) Find a Galois extension $M/\mathbb{Q}$ with Galois group isomorphic to a semidirect product $\mathbb{Z}/5\mathbb{Z} \rtimes (\mathbb{Z}/5\mathbb{Z})^\times$ where $(\mathbb{Z}/5\mathbb{Z})^\times$ acts on $\mathbb{Z}/5\mathbb{Z}$ via multiplication.
- (9) Suppose that $M/\mathbb{Q}$ is a Galois extension of degree 99. Show that there exist subextensions $L_1/\mathbb{Q}$ and $L_2/\mathbb{Q}$ of degrees 9 and 11, respectively, and that they are both Galois over $\mathbb{Q}$.